Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Select the incorrect statement about N_2F_4 and N_2H_4 :
(I) In N_2F_4, d-orbitals are contracted by electronegative fluorine atoms, but d-orbital contraction is not possible by H-atom in N_2H_4.
(II) The $N-N$ bond energy in N_2F_4 is more than $N-N$ bond energy in N_2H_4
(III)The $N-N$ bond length in N_2F_4 is more than that of in N_2H_4
(IV) The $N-N$ bond length in N_2F_4 is more less than that of in N_2H_4 Choose the correct code :

(A) I, II and III

(B) I and III

(C) II and IV

(D) II and III

Q.2

Select pair of compounds in which both have different hybridization but have same molecular geometry.

(A) BF_3, BrF_3

(B) $ICl_2^{\Theta}, BeCl_2$

(C) BCl_3, PCl_3

(D) PCl_3, NCl_3

Q.3

Assuming that all the four valencies of carbon atom in propane pointing towards the corners of regular tetrahedron. Calculate the distance between the terminal carbon atoms in propane. (Given C-C single bond length is 1.54 A°)

(A) $2.514A^{\circ}$

(B) $0.961A^{\circ}$

(C) $5.37A^{\circ}$

(D) $0.268A^{\circ}$

Q.4

Anhydrous $AlCl_3$ is covalent from the data given below. Predict whether it would remain covalent or become ionic in aqueous solution.
Ionization energy for $AlCl_3 = 5137KJ/$ mole.

@IITJEE_Advanced

$\Delta H_{\text{hydration for}}\ Al^{+3}$ is $-4665 KJ/$ mole.
$\Delta H_{\text{hydration}}$ for Cl^{-} is $-381 KJ/$ mole.

(A) covalent

(B) Ionic

(C) neither covalent nor ionic

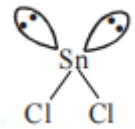
(D) cannot be predicted

Q.5

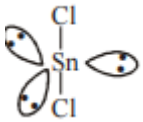
The atomic number of Sn is 50. The shape of gaseous $SnCl_2$ molecule is:

(A) $Cl - Sn - Cl$

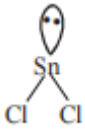
(B)



(C)

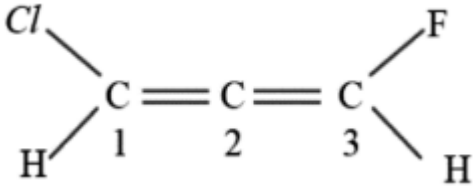


(D)



Q.6

Consider the following molecule



If π-electron cloud of $C_1 - C_2$ is present in the plane of paper then

(A) Fluorine is perpendicular to the plane of paper

(B) Chlorine is present in the plane of paper

(C) σ - bond of $C_2 - C_3$ is perpendicular to the plane of paper

(D) π - electron cloud of $C_2 - C_3$ bond and Cl is present in same plane

Q.7

Assuming Covalent radii to be additive property. Calculate the lodine - lodine distance in ortho and meta diiodo benzene molecules. The benzene is a regular hexagon and $C - I$ lies on a line passing through the centre of the hexagon. The $C - C$ bond length in benzene is $1.4 \mathring{A}$ and covalent radius of I and C atoms are $1.33 \mathring{A}$ and $0.77 \mathring{A}$. Also neglect different overlapping effects.

@IITJEE_Advanced

(A) $3.5A^\circ$ and $6.06A^\circ$

(B) $1.25A^\circ$ and $3.76A^\circ$

(C) $5.43A^\circ$ and $2.86A^\circ$

(D) $1.87A^\circ$ and $3.91A^\circ$

Q.8

In which of the dimerisation process, the achievement of the octet is not the driving force.

(A) $2AlCl_3 \longrightarrow Al_2Cl_6$

(B) $BeCl_2 \longrightarrow BeCl_2$ (solid)

(C) $2ICl_3 \longrightarrow I_2Cl_6$

(D) $2NO_2 \longrightarrow N_2O_4$

Q.9

Which of the following process is/are associated with change of hybridization of the underlined compound ?

(A) $\underline{Al(OH)_3}$ ppt. dissolved in $NaOH$

(B) $\underline{B_2H_6}$ is dissolved in THF

(C) $\underline{SiF_4}$ vapour is passed through liq. HF

(D) Solidification $\underline{PCl_5}$ vapour

Q.10

In the structure of H_2CSF_4, which of the following statement is/are correct -

(A) Two $C-H$ bonds are in the same plane of axial $S-F$ bonds

(B) Two $C-H$ bonds are in the same plane of equitorial $S-F$ bonds

(C) Total six atoms are in the same plane

(D) Equitorial $S-F$ plane is perpendicular to plane of π-bond

Q.11

Choose the correct statements -

(A) The co-ordinate bond is called semi polar bond

(B) The reaction of covalent compound are slow

(C) When two atoms combine to form a molecule energy is released

(D) AlI_3 is more covalent than $AlBr_3$

Q.12

@IITJEE_Advanced

Passage:

Polar covalent molecules exhibit dipole moment. Dipole moment is equal to the product of charge separation, q and the bond length d for the bond. Unit of dipole moment is debye. One debye is equal to 10^{-18} esu cm.

Dipole moment is a vector quantity. It has both magnitude and direction. Hence, dipole moment of a molecule depends upon the relative orientation of the bond dipoles, but not on the polarity of bonds alone. A symmetrical structure shows zero dipole moment. Thus, dipole moment helps to predict the geometry of a molecules. Dipole moment values can be used to distinguish between cisand trans-isomers, ortho-, meta- and para-forms of a substance, etc.

Question:

Which is a polar molecule -

(A) $\mathrm{XeF_4}$

(B) $\mathrm{BF_3}$

(C) $\mathrm{I_2Cl_6}$

(D) $\mathrm{PCl_2\ F_3}$

Q.13

Passage:

Polar covalent molecules exhibit dipole moment. Dipole moment is equal to the product of charge separation, q and the bond length d for the bond. Unit of dipole moment is debye. One debye is equal to 10^{-18} esu cm.

Dipole moment is a vector quantity. It has both magnitude and direction. Hence, dipole moment of a molecule depends upon the relative orientation of the bond dipoles, but not on the polarity of bonds alone. A symmetrical structure shows zero dipole moment. Thus, dipole moment helps to predict the geometry of a molecules. Dipole moment values can be used to distinguish between cisand trans-isomers, ortho-, meta- and para-forms of a substance, etc.

Question:

A diatomic molecule has a dipole moment of $1.2\mathrm{D}$. If the bond length is $1.0 \times 10^{-8}\ \mathrm{cm}$, what fraction of charge does exist each atom ?

(A) 0.1

(B) 0.2

(C) 0.25

(D) 0.3

Q.14

Passage:

Polar covalent molecules exhibit dipole moment. Dipole moment is equal to the product of charge separation, q and the bond length d for the bond. Unit of dipole moment is debye. One debye is equal to 10^{-18} esu cm.

Dipole moment is a vector quantity. It has both magnitude and direction. Hence, dipole moment of a molecule depends upon the relative orientation of the bond dipoles, but not on the polarity of bonds alone. A symmetrical structure shows zero dipole moment. Thus, dipole moment helps to predict the geometry of a molecules. Dipole moment values can be used to distinguish between cisand trans-isomers, ortho-, meta- and para-forms of a substance, etc.

@IITJEE_Advanced

Question:

Arrange the following compounds in increasing order of dipole moments, toluene (I) o-dichlorobenzene (II), m-dichlorobenzene (III) and p-dichlorobenzene (IV).

(A) IV $<$ I $<$ II $<$ III

(B) I $<$ IV $<$ II $<$ III

(C) IV $<$ I $<$ III $<$ II

(D) IV $<$ II $<$ I $<$ III

Q.15

Passage:

The distribution of electrons among various molecular orbitals is called the electronic configuration of the molecule which provides us the following very important information about the molecule.

(A) Stability of molecule : The molecule is stable if number of bonding molecular orbital electrons (N_b) is greater than the number of antibonding molecular orbital electrons (N_s).

(B) Bond order : Bond order $= \frac{1}{2}\left(N_b - N_a\right)$

A positive bond order means a stable molecule while a negative or zero bond order means an unstable molecule.

(C) Nature of the bond: Bond order 1,2 and 3 corresponds to single, double and triple bonds respectively.

(D) Bond length : Bond length decreases as bond order increases.

(E) Magnetic nature : Molecular orbitals in a molecule are doubly occupied, the substance is diamagnetic and if one or more molecular orbitals are singly occupied, it is paramagnetic.

Question:

Which of the following statement is incorrect -

(A) Among O_2^+, O_2 and O_2^- the bond length decreases as $O_2^- > O_2$ and O_2^+

(B) He_2 molecule does not exist as the bonding and anti-bonding orbitals cancel each other

(C) C_2, $O_2{}^{2-}$ and Li_2 are diamagnetic

(D) In F_2 molecule, the energy of σ_{2p_z} is more than π_{2px} and π_{2p_y}

Q.16

Passage:

The distribution of electrons among various molecular orbitals is called the electronic configuration of the molecule which provides us the following very important information about the molecule.

(A) Stability of molecule : The molecule is stable if number of bonding molecular orbital electrons (N_b) is greater than the number of antibonding molecular orbital electrons (N_s).

(B) Bond order : Bond order $= \frac{1}{2}\left(N_b - N_a\right)$

A positive bond order means a stable molecule while a negative or zero bond order means an unstable molecule.

(C) Nature of the bond: Bond order 1,2 and 3 corresponds to single, double and triple bonds respectively.

(D) Bond length : Bond length decreases as bond order increases.

(E) Magnetic nature : Molecular orbitals in a molecule are doubly occupied, the substance is diamagnetic and if one or more molecular orbitals are singly occupied, it is paramagnetic.

Question:

@IITJEE_Advanced

The following molecules/species have been arranged in the order of their increasing bond orders. Identify the correct order : (I) O_2; (II) O_2^- (III) O_2^{2-}; (IV) O_2^+

(A) III $<$ II $<$ I $<$ IV

(B) IV $<$ III $<$ II $<$ I

(C) III $<$ II $<$ IV $<$ I

(D) II $<$ III $<$ I $<$ IV N_2

Q.17

Passage:

The distribution of electrons among various molecular orbitals is called the electronic configuration of the molecule which provides us the following very important information about the molecule.

(A) Stability of molecule : The molecule is stable if number of bonding molecular orbital electrons (N_b) is greater than the number of antibonding molecular orbital electrons (N_s).

(B) Bond order : Bond order $= \frac{1}{2}(N_b - N_a)$

A positive bond order means a stable molecule while a negative or zero bond order means an unstable molecule.

(C) Nature of the bond: Bond order 1,2 and 3 corresponds to single, double and triple bonds respectively.

(D) Bond length : Bond length decreases as bond order increases.

(E) Magnetic nature : Molecular orbitals in a molecule are doubly occupied, the substance is diamagnetic and if one or more molecular orbitals are singly occupied, it is paramagnetic.

Question:

N_2 has greater dissociation energy than N_2^+, where as O_2 has a lower dissociation energ because -

(A) bond order is reduced when O_2 is ionized to O_2^+ and bond order is increased ionized to N_2^+

(B) bond order is increased when O_2 is ionized to O_2^+ and bond order is decreased ionized to N_2^+

(C) bond order is decreased when O_2 is ionized to O_2^+ and bond order is increased ionized to N_2^+

(D) None of these

Q.18

Match the column -

Column I	**Column II**
(A) NH_4Cl	(p) Hydrogen bond
(B) $CuSO_4.5H_2O$	(q) Co-ordinate bond
(C) HNC	(r) Ionic bond
(D) Liquid H_2O_2	(s) Covalent bond

(A) A-q,r,s; B-p,q,r,s ; C- q,s ; D-p,s;

(B) A-p,q; B-r ; C- p,q,s ; D-p,q,s;

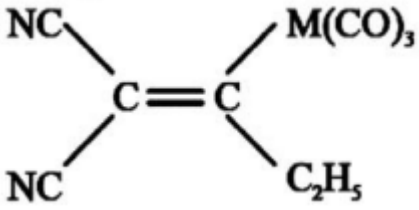
(C) A-p,q,s; B-p,r ; C- p,q,r,s ; D-p,q,s;

(D) A-r,s; B-p,r ; C- p ; D-p,q,s;

@IITJEE_Advanced

Q.19

Total number of σ and π bonds present in following structure is $6x$ then find the value of x?



Q.20

The hybridization of central atoms of compounds A,B,C and D are sp^3d, sp^3, sp^2 & sp respectively. If compounds A and D have the same shape like I_3^- and compounds B and C have the same shape like water molecule.Then calculate value of $P+Q+R+S$ where P, Q, R and S are number of lone pairs on central atoms of compounds A, B, C and D respectevely.

Answers & Solutions

Q.1 Answer:

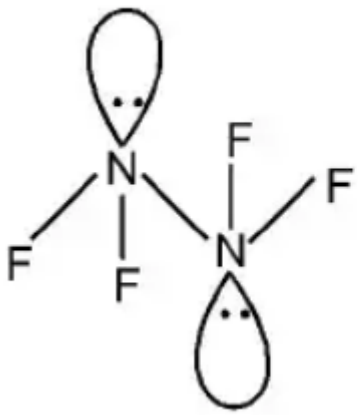
I and III

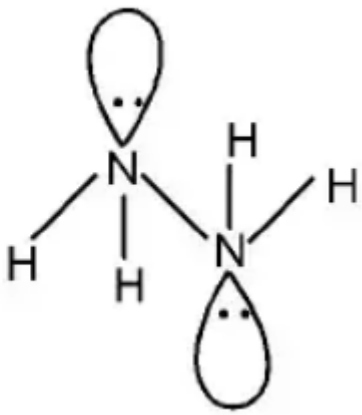
Solution:

1) s-character tend to concentrate in orbitals that are directed toward electropositive groups and p-character tends to concentrate in orbitals that are directed toward electronegative groups.

2) It can be explained on the basis of Bent's rule. In N_2F_4, $N-N$ bond has more s-character, hence, bond length decreases.

(3,4) While in N_2H_4, $N-N$ bond has less s-character (i.e. more p-character), hence bond length increases.





Q.2 Answer:

$ICl_2^{\Theta}, BeCl_2$

Solution:

BF_3 sp^2 **Trigonal planar**

BrF_3 sp^3d **Bent 'T' shape**

$ICl_2\Theta$ sp^3d **Linear**

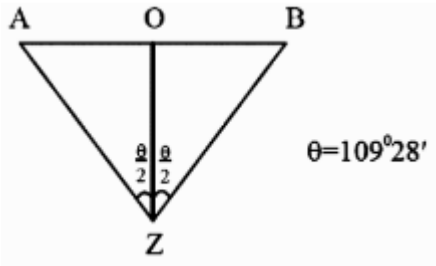
@IITJEE_Advanced

$BeCl_2$ sp Linear
BCl_3 sp^2 Trigonal planar
PCl_3 sp^3 **Pyramidal**
PCl_3 sp^3 **Pyramidal**
NCl_3 sp^3 **Pyramidal**

Q.3 Answer:

$2.514A^\circ$

Solution:



$AB = 2 \times AO \quad \frac{AO}{AZ} = \sin\frac{\theta}{2}$
$AO = AZ \times \sin\frac{\theta}{2} = 1.54 \times \sin 54°44' = 1.257$
$(AZ = C - C$ **bond length which is** $1.54\mathring{A})$
$AB = 2AO = 2 \times 1.257 = 2.514\mathring{A}$

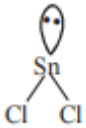
Q.4 Answer:

Ionic

Solution:

When hydration energy is greater than ionization energy, the bond is ionic. When Ionization energy is greater than hydration energy the bond is covalent.
Hydration energy: $-4665 - 3 \times 381KJ$
Ionization energy: $+5137KJ$
$\therefore$ **Hydration energy** $+$ **Ionization energy** $= -671$ KJ
Hydration energy is greater than Ionization energy
$\therefore$ $AlCl_3$ **becomes ionic in aqueous solution.**

Q.5 Answer:

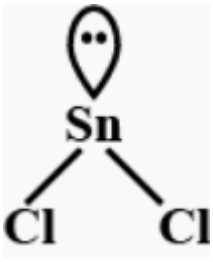


Solution:

The atomic number of Sn **is 50 . The shape of gaseous** $SnCl_2$ **molecule is angular as shown by option** D**. This is due to the presence of one lone pair of electron and two bond pairs of electrons which result in** sp^2**.**

@IITJEE_Advanced

hybridisation.



Q.6 Answer:

π - electron cloud of $C_2 - C_3$ bond and Cl is present in same plane

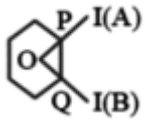
Solution:

$(C_1 - C_2)\pi$-bond is perpendicular to $(C_2 - C_3)\pi$-bond
$(C_1 - C_2)\pi$-bond is also perpendicular to $(C_1 - Cl)$ bond
So $(C_2 - C_3)\pi$ - electron cloud and $(C_2 - Cl)$ bond is in same plane.

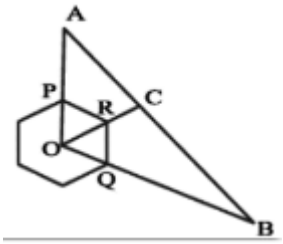
Q.7 Answer:

$3.5A^\circ$ and $6.06A^\circ$

Solution:



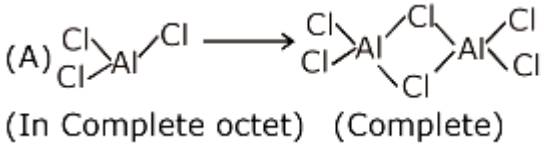
$\mathrm{OQ} = \mathrm{OP} = \mathrm{C} - \mathrm{C}$ **bond length**
$\mathrm{QB} = \mathrm{PA} = \mathrm{C} - \mathrm{I}$ **bond length**
$\mathrm{AB} = \mathrm{I} - \mathrm{I}$ **distance** $= \mathrm{OP} + \mathrm{PA} = \mathrm{OP} + \mathrm{r_c} + \mathrm{r_l}$
$= 1.4 + 1.33 + 0.77 = 3.5^\circ\mathrm{A}$
($r_\mathrm{C} =$ **radius of carbon atom,** $\mathrm{r_1} =$ **radius of iodine atom)**



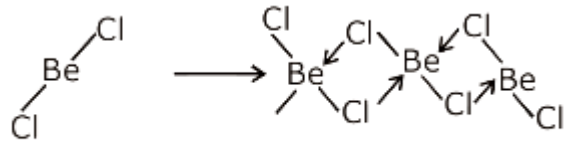
Q.8 Answer:

$2\mathrm{ICl_3} \longrightarrow \mathrm{I_2Cl_6}$

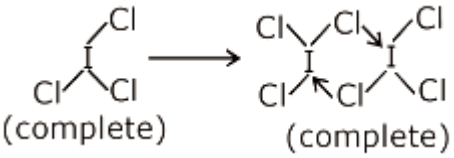
Solution:



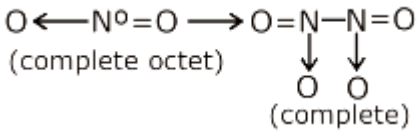
@IITJEE_Advanced

(B) 

(In compte octet) Complete octet

(C) 

(complete) (complete)

(D) 

(complete octet) (complete)

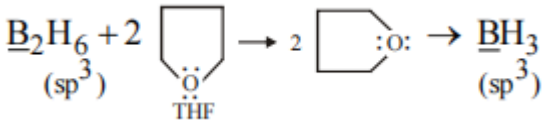
Q.9 Answer:

$\underline{Al(OH)_3}$ ppt. dissolved in $NaOH$

Solution:

(A) $\underline{Al}(OH)_3 + NaOH \longrightarrow Na\,[\underline{Al}(OH)_4]$

(B)



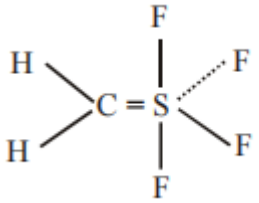
(C) $\underset{sp^3\,d}{SiF_4} + 2HF \longrightarrow \underset{sp^3\,d^2}{H_2\,SiF_6}$

(D) PCl_5 **(g)** $\rightarrow PCl_5(\,s)$

Q.10 Answer:

Two $C-H$ bonds are in the same plane of axial $S-F$ bonds

Solution:



Hydrogen atoms are in a vertical plane with axial fluorine atoms π-bond involving a p-orbital of carbon atom must lie in equatorial plane of the molecule.

Six atoms, i.e., $2H$-atoms, C, S and both axial F-atoms lie in one plane.

@IITJEE_Advanced

Q.11 Answer:

The co-ordinate bond is called semi polar bond

Solution:

The co-ordinate bond is called semi polar bond.

The reaction of covalent compound are slow in comparison to ionic compound.

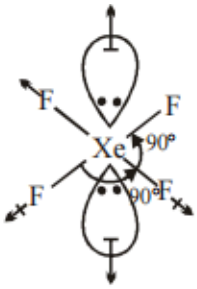
When two atoms combine to form a molecule energy is released.

AlI_3 **is more covalent than** $AlBr_3$.

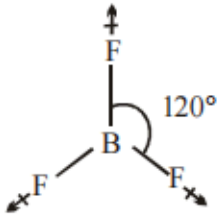
Q.12 Answer:

PCl_2F_3

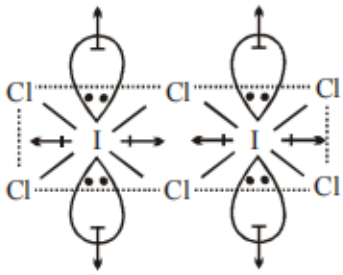
Solution:



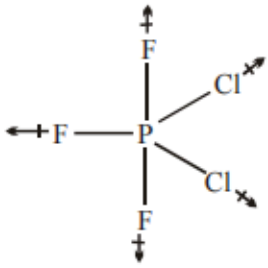
Non-polar
$\mu = 0$
(a)



Non-polar
$\mu = 0$
(b)



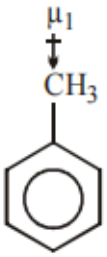
Planar structure
$\mu = 0$
(c)



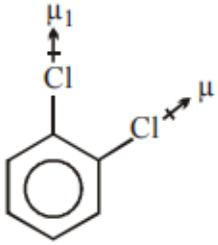
$m \neq 0$
(d)

(ii) Fraction of charge $= \dfrac{1.2 \times 10^{-18}}{4.8 \times 10^{-10} \times 10^{-8}} = 0.25$

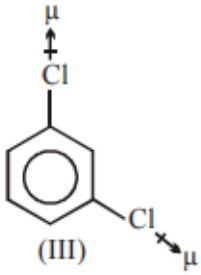
(iii)



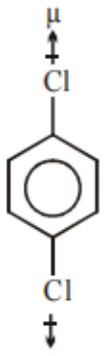
(I)
μ_1
$\mu > \mu_1$



(II)
2μ



(III)
μ



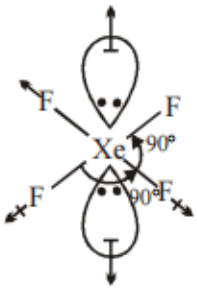
(IV)
$\mu = 0$

Q.13 Answer:

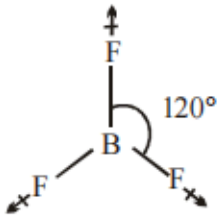
0.25

Solution:

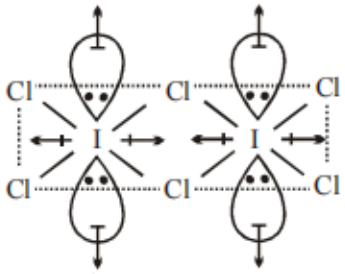
@IITJEE_Advanced



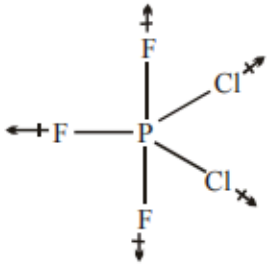
Non-polar
$\mu = 0$
(a)



Non-polar
$\mu = 0$
(b)



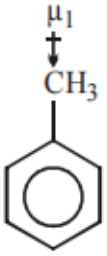
Planar structure
$\mu = 0$
(c)



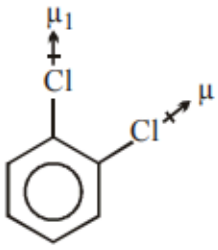
$m \neq 0$
(d)

(ii) Fraction of charge $= \dfrac{1.2 \times 10^{-18}}{4.8 \times 10^{-10} \times 10^{-8}} = 0.25$

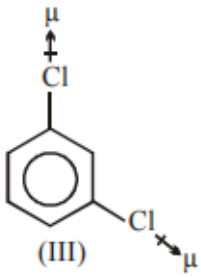
(iii)



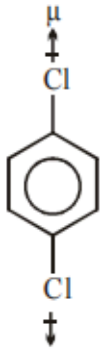
(I)
μ_1
$\mu > \mu_1$



(II)
2μ



(III)
μ



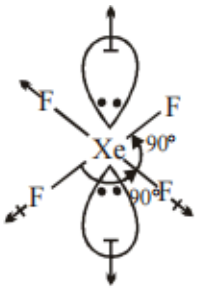
(IV)
$\mu = 0$

Q.14 Answer:

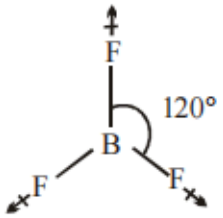
IV $<$ I $<$ III $<$ II

Solution:

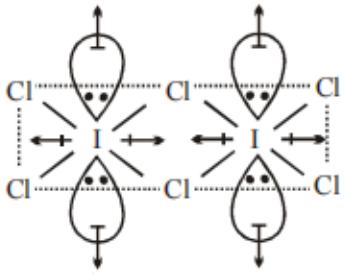
@IITJEE_Advanced



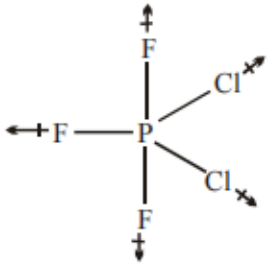
Non-polar
$\mu = 0$
(a)



Non-polar
$\mu = 0$
(b)



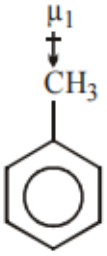
Planar structure
$\mu = 0$
(c)



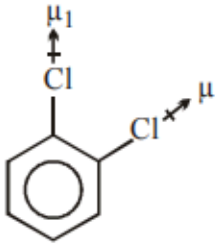
$m \neq 0$
(d)

(ii) Fraction of charge $= \dfrac{1.2 \times 10^{-18}}{4.8 \times 10^{-10} \times 10^{-8}} = 0.25$

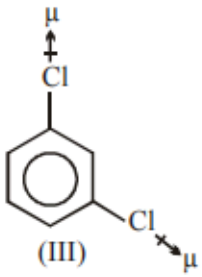
(iii)



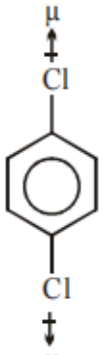
(I)
μ_1
$\mu > \mu_1$



(II)
2μ



(III)
μ



(IV)
$\mu = 0$

Q.15 Answer:

In F_2 molecule, the energy of σ_{2p_z} is more than π_{2px} and π_{2p_y}

Solution:

(A) Bond order $\propto 1/$ bond length

Bond order $O_2^+ = 2.5, O_2 = 2, O_2^- = 1.5$

So correct order of bond length is $O_2^- > O_2 > O_2^+$

(B) Bond order of $He_2 = \frac{2-2}{2} = 0$**. So** He_2 **does not exist**

(C) In all these molecules all electrons are paired in molecular orbitals. So they are all diamagnetic.

(D) -Molecular orbital electronic configuration is given by,

MOEC $=$

$$KK(\sigma 2s)^2 (\sigma^* 2s)^2 (\sigma 2p_z)^2 \left[(\pi 2p_x)^2 = (\pi 2p_y)^2\right] \left[(\pi^* 2p_x)^2 = (\pi^* 2p_y)^2\right]$$

-where, KK denotes $(\sigma 1s)^2 (\sigma^* 1s)^2$

Q.16 Answer:

$III < II < I < IV$

Solution:

	O_2	O_2^-	O_2^{2-}	O_2^+
Bond order	2	1.5	1	2.5

Q.17 Answer:

@IITJEE_Advanced

bond order is increased when O_2 is ionized to O_2^+ and bond order is decreased ionized to N_2^+

Solution:

Species	N_2	N_2^+	O_2	O_2^+
Bond order	3	2.5	2	2.5

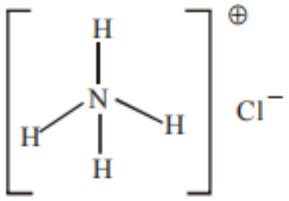
Q.18 Answer:

A-q,r,s; B-p,q,r,s ; C- q,s ; D-p,s;

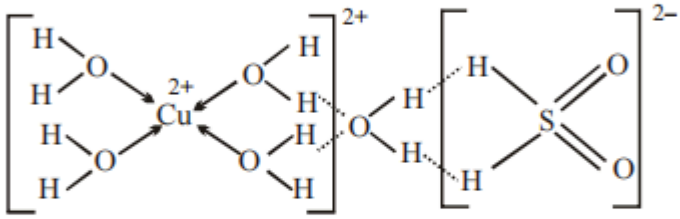
Solution:

((A) - qrs, (B) - pqrs, (C) qs, (D) - ps)

(A) NH_4Cl :



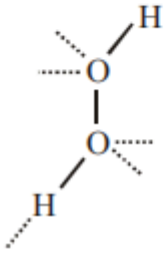
(B) $CuSO_4.5H_2O$:



(C) HNC :

$H-N\equiv C$

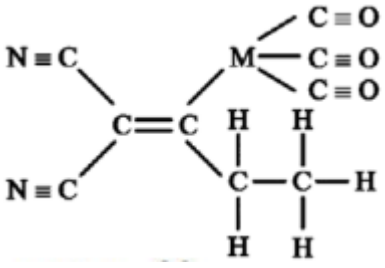
(D) Liquid H_2O_2 :



Q.19 Answer:

—

Solution:



$\therefore \sigma + \pi = 30$

Q.20 Answer:

—

@IITJEE_Advanced

Solution:

$A \rightarrow Sp^3d \rightarrow 3LP \rightarrow$ **linear**

$B \rightarrow Sp^3 \rightarrow 2LP \rightarrow$ **angular**

$C \rightarrow Sp^2 \rightarrow 1LP \rightarrow$ **angular**

$D \rightarrow Sp \rightarrow 0LP \rightarrow$ **linear**

$P + Q + R + S = 3 + 2 + 1 + 0 = 6$

@IITJEE_Advanced